Fuzzy Sets, Logic and Systems & Applications

Week 1 : Assignment 1 Solution

Q.1. What is the membership value for a triangular membership function with parameters a=2, b=6, and c=10 when x=8?

(a) 1.0

(b) 0.5

(c) 0.0

(d) None of the above

Solution: (b)

$$triangle(x; a, b, c) = \max\left\{\min\left[\frac{x-a}{b-a}, \frac{c-x}{c-b}\right], 0\right\}$$

$$triangle(x; a, b, c) = \max\left\{\min\left[\frac{8-2}{6-2}, \frac{10-8}{10-6}\right], 0\right\} = \max\left\{\min\left[\frac{6}{4}, \frac{2}{4}\right], 0\right\} = \max\{\min[1.5, 0.5], 0\} = \max\{0.5, 0\}$$

$$triangle(x; a, b, c) = 0.5$$

Q.2. For a triangular membership function with parameters a=2, b=6, and c=10, find the value of x for which the membership value is maximized.

(a) 6

(b) 2

(c) 10

(d) None of the above

Solution: (a)

$$triangle(x; a, b, c) = \max\left\{\min\left[\frac{x-a}{b-a}, \frac{c-x}{c-b}\right], 0\right\}$$

$$triangle(x; a, b, c) = \max\left\{\min\left[\frac{6-2}{6-2}, \frac{10-6}{10-6}\right], 0\right\} = \max\{\min[1, 1], 0\} = \max\{1, 0\} = 1.0$$

Q.3. For a triangular membership function with parameters a=2, b=6, and c=10, if b is changed to 8, which of the following statements are true:

(a) At x=4 and x=8, the degree of membership will be 0.5

(b) At x=5 and x=9, the degree of membership will be 0.5

(c) At x=6 and x=8, the degree of membership will be 0.5

(d) None of the above

Solution: (b)

For new parameters i.e., a=2, b=8, and c=10, the following are the membership values,

$$triangle(x=5; a=2, b=8, c=10) = \max\left\{\min\left[\frac{5-2}{8-2}, \frac{10-5}{10-8}\right], 0\right\} = \max\{\min[0.5,2.5], 0\}$$
$$= \max\{0.5,0\} = 0.5$$

$$triangle(x=9; a=2, b=8, c=10) = \max\left\{\min\left[\frac{9-2}{8-2}, \frac{10-9}{10-8}\right], 0\right\} = \max\left\{\min\left[\frac{7}{6}, 0.5\right], 0\right\}$$
$$= \max\{0.5,0\} = 0.5$$

Q.4. What is the membership value for a trapezoidal membership function with parameters a=1, b=3, c=7, and d=9 when x=7?

(a) 1.0

(b) 0.5

(c) 0.0

(d) None of the above

Solution: (a)

$$trapezoid(x; a, b, c, d) = \max\left\{\min\left[\frac{x-a}{b-a}, 1, \frac{d-x}{d-c}\right], 0\right\}$$

$$trapezoid(x; a, b, c, d) = \max\left\{\min\left[\frac{7-1}{3-1}, 1, \frac{9-7}{9-7}\right], 0\right\} = \max\{\min[3,1,1], 0\} = \max\{1,0\} = 1.0$$

Q.5. For a trapezoidal membership function with parameters a=1, b=3, c=7, and d=9, find the values of x for which the membership value is maximized.

(a) x≤1

(b) x≥9

(c) 3≤x≤7

(d) None of the above

Solution: (c)

Q.6. For a trapezoidal membership function with parameters a=1, b=3, c=7, and d=9,if b is changed to 5, which of the following statements are true:

(a) At x=2 and x=8, the degree of membership will be 0.5

(b) At x=3 and x=8, the degree of membership will be 0.5

(c) At x=5 and x=8, the degree of membership will be 0.5

(d) None of the above

Solution: (b)

$$trapezoid(x; a, b, c, d) = \max\left\{\min\left[\frac{x-a}{b-a}, 1, \frac{d-x}{d-c}\right], 0\right\}$$

$$trapezoid(x=3; a=1, b=5, c=7, d=9) = \max\left\{\min\left[\frac{3-1}{5-1}, 1, \frac{9-3}{9-7}\right], 0\right\} = \max\{\min[0.50,1,3.0], 0\} = 0.5$$

$$trapezoid(x = 8; a = 1, b = 5, c = 7, d = 9) = \max\left\{\min\left[\frac{8-1}{5-1}, 1, \frac{9-8}{9-7}\right], 0\right\} = \max\{\min[1.75,1,0.5], 0\} = 0.5$$

Q.7. Which of the following is the correct expression for a Gaussian membership function, if the parameters are specified by $[\mu, \sigma]$?

(a) $e^{-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2}$

(b) $e^{-\left(\frac{(x-\mu)^2}{2\sigma^2}\right)}$

(c) $e^{-\left(0.5\times\left(\frac{x-\mu}{\sigma^2}\right)^2\right)}$

(d) None of the above

Solution: (a) and (b)

Q.8. How does the width of a Gaussian membership function change with an increase in the value of sigma (σ)?

(a) Increases

(b) Decreases

(c) Remains constant

(d) Unpredictable or not determined by sigma

Solution: (a)

Q.9. For a generalized bell shaped membership function specified by three parameters [a,b,c], which of the following statements are true?

(a) Parameter a defines the shape of the curve

(b) Parameter a defines the width of the membership function

(c) Parameter a defines the centre of the membership function

(d) None of the above

Solution: (b)

Q.10. For a generalized bell shaped membership function specified by three parameters [a,b,c], which of the following statements are true?

(a) Larger value of b creates a wider membership function

(b) Larger value of b creates steeper transition

(c) Parameter b defines the shape of the curve on either side of the central plateau

(d) None of the above

Solution: (b) and (c)